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ORIGINAL ARTICLE

A multi-UAV rapid post-disaster search and rescue method based on deep reinforcement learning

Li Tan[1] · Haixia Zhao[1]

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Abstract

Deep reinforcement learning shows broad prospects in multi-unmanned aerial vehicle(UAV) collaborative search and rescue tasks. However, in the face of high-dimensional collaborative decision-making spaces and limited computing resources, its performance is vulnerable to limitations. This paper proposes a deep deterministic policy gradient method based on linear attention. By introducing the linear attention mechanism based on random feature mapping, while effectively modeling the interaction among UAVs, the computational and storage overcosts caused by the increase in the number of UAVs have been significantly reduced. Furthermore, by combining smooth experience replay and adaptive importance sampling mechanism, the training efficiency and strategy stability have been further improved. The simulation experiments on both post-disaster response search and dynamic containment tasks demonstrate that the proposed algorithm consistently outperforms existing methods. In small-scale scenarios, it maintains nearly perfect success rates, while in medium- and large-scale settings it achieves up to 90.6% and 85.2% success rates in the post-disaster response search task and up to 90.1% and 80.2% in the containment task, corresponding to relative improvements of 15–21% over baselines. These results highlight both the robustness of the method in simple cases and its clear advantage under more challenging multi-UAV conditions.

Keywords Multi-UAV systems · Deep reinforcement learning · Random feature attention · Prioritized experience replay

Introduction

The early stage of a disaster is a critical period for personnel search and rescue (SAR), where rapid and accurate localization of lost persons is crucial for improving rescue success rates [1]. However, complex terrain, paralyzed infrastructure, and hazardous environments often limit traditional SAR approaches [2]. In recent years, unmanned aerial vehicle (UAV) technology has become increasingly integral to post-disaster SAR, gradually evolving from a single auxiliary tool to an intelligent platform with active sensing and collaborative decision-making capabilities [3]. Multi-UAV collaborative search, in particular, leverages efficiency and flexibility to meet the demands of wide coverage and rapid response, supported by recent advances in UAV detection [4], making it a vital approach for enhancing SAR efficiency [5]. Nevertheless, real-world SAR missions face significant challenges, including the generation of numerous suspected targets by life detectors or cell phone signals, often requiring advanced object detection methods [6], which overwhelm single or small-scale UAV systems, and the unpredictable behavioral paths of lost or escaped individuals, necessitating dynamic monitoring and real-time adaptability. These complexities drive the development of Multi-UAV Systems towards larger scale and higher intelligence [7–9]. While increasing the number of UAVs enhances search coverage, it also brings a rapid rise in system complexity, particularly due to the lack of lightweight designs for efficient multi-UAV cooperation. Existing collaboration strategies often rely on high-frequency communication and centralized control [10], which struggle to maintain real-time performance and robustness in post-disaster scenarios with limited communication and scarce computational resources, leaving the need for computationally efficient and robust cooperation mechanisms unaddressed. Consequently, developing distributed, robust, and computationally efficient cooperative mechanisms remains a critical research

✉ Li Tan
tanli@th.btbu.edu.cn

Haixia Zhao
2330702046@st.btbu.edu.cn

[1] School of Computer and Artificial Intelligence, Beijing Technology and Business University, Beijing, China

priority, with current methods falling short in addressing these efficiency and adaptability challenges, creating a significant research gap [11]. Multi-Agent Reinforcement Learning (MARL) shows promise for multi-UAV cooperative control. Unlike traditional optimization algorithms (e.g., genetic algorithms) or rule-based control, MARL dynamically adapts to high-uncertainty environments, supports collaborative decision-making, and enables distributed execution in communication-constrained scenarios [12, 13]. Among MARL methods, Multi-Agent Deep Deterministic Policy Gradient (MADDPG) is widely adopted due to its support for continuous action spaces and flexible environment modeling [14]. MADDPG enhances synergy through a centralized training and distributed execution framework. However, as the number of UAVs increases, its centralized critic structure must model all agent states and actions, leading to a significant computational burden, with efficient alternatives still underexplored. Additionally, in post-disaster environments with high interference and restricted communication, MADDPG struggles to extract critical information, resulting in reduced collaborative efficiency and highlighting the need for methods that ensure robustness under such constraints. To address these limitations, some studies have introduced attention mechanisms, such as the Multi-Agent Attention Critic (MAAC), which dynamically weights key agent states using softmax attention to improve decision robustness [15]. However, MAAC's softmax attention is computationally complex, and its training and inference speeds degrade significantly with large numbers of UAVs, making it poorly suited for time-critical post-disaster scenarios. This underscores a critical gap in achieving computationally efficient and robust multi-UAV collaboration under resource constraints.

To bridge this research gap, this paper proposes an improved MADDPG method based on linear attention mechanism (LADDPG). The method utilizes the linear attention structure to selectively aggregate key intelligent states, significantly reducing the computational burden while improving the information fusion efficiency [16]. The main contributions of this paper are as follows:

(1) Aiming at the problem of limited system scale expansion and low training efficiency in multi-UAV collaborative search and rescue missions, a linear attention mechanism based on stochastic feature mapping is designed, which utilizes the kernel function approximation to replace the softmax operation, significantly reduces the computational complexity and improves the training efficiency and system scalability while maintaining the key feature selection capability.
(2) To improve the stability and robustness of policy training, we propose an adaptively enhanced prioritized experience playback mechanism, which effectively mitigates the learning instability triggered by noisy samples in post-disaster scenarios by smoothing the temporal difference (TD) error and dynamically adjusting the sampling weights.
(3) In the simulation of post-disaster response search and dynamic target fencing and control tasks, multi-UAV simulation experiments of different scales are constructed to compare the performance of the proposed method with mainstream policy learning algorithms in terms of collaborative efficiency and system stability, which demonstrates good response speed and system scalability.

The rest of this paper is organized as follows. In Sect. Related work, related work is reviewed. The proposed LADDPG is introduced in Sect. Systemmodel. The performance evaluation and results are presented in Sect. Experiment, and finally, this paper is concluded in Sect. Conclusion.

Related work

In complex post-disaster environments, multi-UAVs are widely used in large-scale search and rescue missions due to their rapid deployment, high mobility and distributed operational capabilities. Recent advances include trajectory forecasting using Long Short-Term Memory (LSTM) neural networks to predict dynamic target paths [17] and signals intelligence-based detection to distinguish UAV control signals [18], enhancing perception for collaborative tasks. Collaborative search missions usually involve large target distribution and multi-target uncertainty, which makes how UAV clusters efficiently assign tasks and dynamically plan paths a research priority [19, 20]. To this end, several distributed and centralized control methods have been proposed. Xing et al. [21] achieved task allocation optimization through area division; Xiong et al. [22] designed a multi-intelligence reinforcement learning algorithm based on coalitional games to optimize collaborative strategies; He et al. [23] used graph theory to model collaborative relationships, which effectively improved the synergy effect; and Wang et al. [24] introduced an auction mechanism for task bidding and coordination. However, these rule-based or heuristic strategies have limited performance in scenarios with high target dynamics or drastic environmental changes, and lack the ability to model uncertainty and learn autonomous strategies.

In order to enhance the adaptability of the system to dynamic environments, Deep Reinforcement Learning (DRL) methods have been widely introduced into multi-UAV collaborative tasks. By interacting with the environment to learn strategies, UAVs are able to spontaneously explore the optimal action sequences in task execution.Lowe et al. [14] proposed MADDPG, which applies centralized training and

decentralized execution to multi-agent decision making, and becomes a representative algorithm in multi-UAV control; Xiao et al. [25] propose asynchronous Actor-Critic algorithm to improve the efficiency of distributed training in MARL. Wang et al. [26] investigated the application of hierarchical reinforcement learning in multi-objective tasks, which effectively improves the task decomposition and planning ability; and Yao et al. [27]proposed a distributed temporal co-consistency algorithm to solve the multi-constraint path planning problem for multi-agents intelligences to reach a goal simultaneously in a dynamic environment. However, with the increase in the number of agents, traditional DRL methods face challenges in information sharing, state modeling, and computational overhead, especially in the context of limited communication and lack of computational resources after a disaster, the efficiency and stability of policy training are difficult to guarantee.

In order to alleviate the state dimension inflation problem in large-scale collaboration, the attention mechanism is introduced into reinforcement learning architectures to enhance the information aggregation efficiency and key state modeling capability.Iqbal and Sha [15] proposed MAAC based on MADDPG. The softmax attention mechanism dynamically focuses on other agent states that are most relevant to the current agent's strategy, which significantly improves the efficiency of multi-agent collaboration; Hu et al. [28] applied the transformer architecture to multi-agent strategy representation, which improved the long-term dependency modeling capability;Feng et al. [29] proposed an algorithm combining graph attention and MARL to improve collaboration efficiency by dynamically modeling inter-agent relationships; Yin et al. [30] combined attention-enhanced inverse reinforcement learning with graph convolution to optimize policy generalizability through implicit reward inference. Ryu et al. [31] further introduced a hierarchical graph attention mechanism to hierarchically model the dependency structure between agents in the actor-critic framework, thereby obtaining a more stable strategy expression in complex collaborative tasks. Wen et al. [32] transformed the MARL problem into sequence modeling and built a Multi-Agent Transformer architecture based on Transformer to capture temporal policy dependencies. Du et al. [33] incorporated heterogeneous graph attention networks in MARL to improve collaboration efficiency by dynamically modeling heterogeneous relationships among agents. In addition, Wu et al. [34] proposed the MADDPG method that fuses the attention mechanism with triple adaptive accuracy to solve the challenge of high-precision cooperative formation control of hybrid heterogeneous UAVs in complex dynamic environments.

In order to further enhance the efficiency of information aggregation and collaboration, researchers have proposed a variety of improvements and variants of the attention mechanism. Sun et al. [35] proposed a multi-agent learning method based on sparse attention mechanism, which significantly reduces the computational complexity by limiting the range of attention computation; Li et al. [36] combined graph convolutional network and attention mechanism to realize spatio-temporal modeling of multi-intelligence dynamics and capture complex interactions under environmental changes; Li et al. [37] proposed a distributed attention mechanism, which reduces the communication dependency and enhances the robustness of the system under communication-constrained environment. However, most of the work still adopts the traditional softmax attention mechanism, whose computational complexity grows quadratically with the number of agents, resulting in high training and reasoning overheads in large-scale collaborative scenarios, limiting its application capabilities. Therefore, designing an efficient attention mechanism with good generalization ability is still an urgent problem in the field of multi-agent reinforcement learning.

System model

Background

In a multi-UAV system, the collaborative decision-making problem is modeled as a partially observable Markov decision process (POMDP), defined by the tuple $(\mathcal{O}, \mathcal{A}, T, R, \gamma)$. Here $\mathcal{O} = \{o_i\}_{i=1}^{N}$ denotes the local observation set. Each observation o_i consists of the UAV's own position and velocity, the relative positions and velocities of other UAVs with respect to it, as well as the relative positions of landmarks, mission targets, and surrounding obstacles. $\mathcal{A} = \{a_i\}_{i=1}^{N}$ represents the joint action space. Each action a_i is a two-dimensional continuous control input, defined as the accelerations in the x and y directions, which determine the UAV's movement direction and magnitude. $T : \mathcal{O} \times \mathcal{A} \to \mathcal{P}(\mathcal{O})$ denotes the state transition function, which specifies the conditional probability distribution of the next observation given the current observation and joint action, thereby capturing the dynamics of the environment under partial observability. $R : \mathcal{O} \times \mathcal{A} \to \mathbb{R}$ is the global reward function, which consists of four parts: a negative distance reward that encourages UAVs to approach mission targets, a positive reward for successfully accomplishing the task, a penalty for collisions with obstacles, and a penalty for collisions with other UAVs. And $\gamma \in [0, 1]$ is the discount factor, which balances immediate rewards against long-term returns during training.

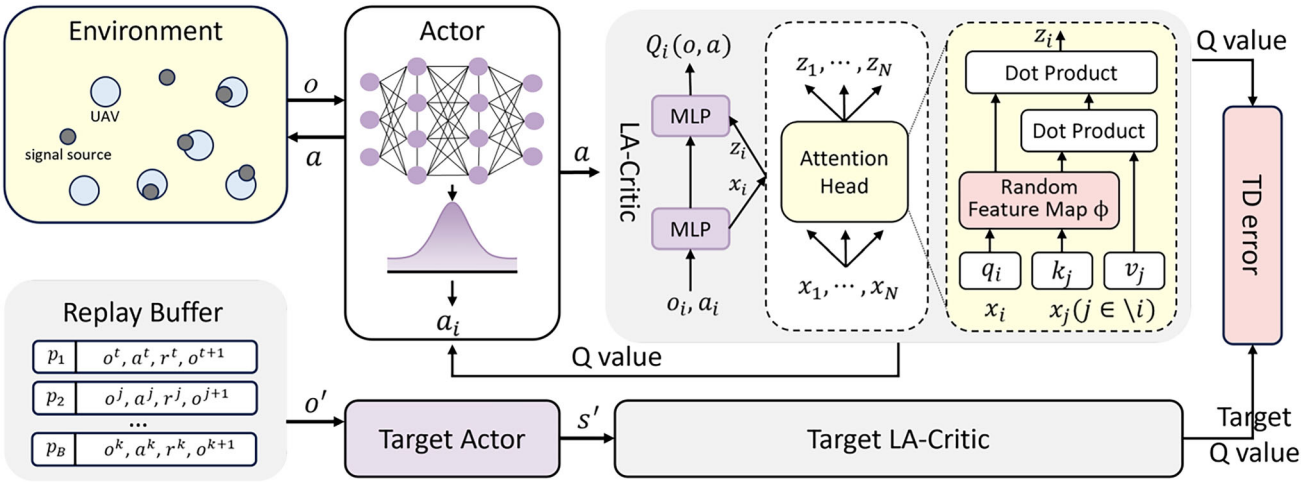


Fig. 1 Schematic view of the LADDPG framework, where o^t denotes the observation at time t, a^t the action, r^t the reward, o^{t+1} the next observation, x_i the encoded state, z_i the attention output, and Q_i the Q-value

Model

In post-disaster search and rescue missions, UAVs operate in highly dynamic and partially observable environments, where collaboration and adaptability are essential. To address these challenges, this work adopts the centralized training and decentralized execution (CTDE) paradigm and applies the MADDPG framework. Under this paradigm, each UAV executes actions based solely on its local observation, while during training a centralized critic leverages joint observations and actions to guide learning. This framework enables UAVs to achieve efficient cooperation while maintaining scalability to large teams.

Each UAV i adopts an actor policy $\pi_i(a_i \mid o_i)$ that maps its local observation to an action, while the critic function $Q_i(o, a)$ evaluates the joint action given the joint observations. Training relies on transitions sampled from a shared replay buffer, and the critic is optimized against the target values $y_i = r_i + \gamma Q_i(o', a'; \theta^-)$, where a' are the next-step actions generated by the target policies and θ^- denotes the parameters of the target critic network, which is softly updated to stabilize learning. As shown in Eq. (1), the critic network parameters θ are trained by minimizing the mean squared TD error.

$$L(\theta) = \mathbb{E}\left[(y_i - Q_i(o, a; \theta))^2\right] \tag{1}$$

where $o = (o_1, \ldots, o_N)$ and $a = (a_1, \ldots, a_N)$ denote the observations and actions of all UAVs. As the number of UAVs increases, the input dimension of the critic network grows linearly, and if interactions between UAVs are modeled with fully connected layers, both parameter size and computational cost scale quadratically, which is impractical for real-time rescue operations.

To alleviate this issue, we design an efficient interaction modeling mechanism based on attention (Fig. 1). Specifically, the local state-action pair of UAV i is first encoded as $x_i = h_i(o_i, a_i)$ through a shared encoding network. Let j index the other UAVs in the system (i.e., $j \neq i$). The contextual representation of UAV i, as defined in Eq. (2), is computed using an attention mechanism.

$$z_i = Attention(q_i, \{k_j, v_j\}) \tag{2}$$

where q_i, k_j, and v_j are the query, key, and value vectors derived from the encoded features. To improve computational efficiency and scalability, we employ random feature mapping to approximate the attention kernel. With a fixed random matrix $R \in \mathbb{R}^{D \times d}$, the feature mapping is defined, as show in Eq. (3).

$$\phi(x) = \frac{1}{\sqrt{2D}}[\sin(Rx), \cos(Rx)] \tag{3}$$

This feature mapping enables the attention output for UAV i to be computed according to Eq. (4).

$$z_i = \frac{\phi(q_i)^\top \sum_{j \neq i} \phi(k_j) v_j^\top}{\phi(q_i)^\top \sum_{j \neq i} \phi(k_j)} \tag{4}$$

This formulation avoids explicitly constructing a dense interaction matrix across all UAVs. It achieves linear time complexity $\mathcal{O}(NDd)$ compared with the traditional quadratic $\mathcal{O}(N^2 d)$, while also reducing memory consumption, thus supporting large-scale deployments. Here, N denotes the number of UAVs, d is the dimension of the encoded features, and D is the number of columns used in the random feature mapping.

The final Q-value for UAV i is obtained from both its encoded state-action feature and the attention-based contextual representation according to Eq. (5).

$$Q_i = F_i(x_i, z_i) = F_i(h_i(o_i, a_i), z_i) \tag{5}$$

where F_i is a two-layer multilayer perceptron that outputs the estimated action value. This structure allows UAVs to incorporate cooperative information without incurring prohibitive computational costs, thereby enhancing collaborative efficiency in disaster-response missions.

In addition to scalable interaction modeling, we enhance the stability of learning by improving prioritized experience replay. In deep reinforcement learning, experiences with larger TD errors contain more information, but conventional prioritized replay schemes that rely solely on instantaneous TD error may be overly sensitive to noise. We introduce an exponential moving average to smooth TD errors, as shown in Eq. (6).

$$\overline{\delta}_i = \lambda \cdot |\delta_i| + (1-\lambda) \cdot \overline{\delta}_{i-1} \tag{6}$$

where $\lambda \in (0, 1)$ controls smoothing. The sampling priority is then defined as $p_i \propto (\overline{\delta}_i)^{\beta}$, which reduces instability caused by transient fluctuations and ensures consistent learning focus.

Furthermore, we propose an adaptive importance-sampling factor σ_t to better balance bias correction during training. Unlike traditional linearly increasing schedules, σ_t is dynamically adjusted based on the variance of recent TD errors, defined by Eq. (7).

$$\sigma_t = \sigma_{t-1} + \eta \cdot (1 - \mathrm{Var}_{\mathrm{TD}}) \tag{7}$$

where Var_{TD} is the normalized variance of TD errors and η is the adjustment rate. This mechanism ensures that $\sigma_t \leq 1.0$, preventing overcorrection and stabilizing training dynamics.

By integrating random feature attention for scalable interaction modeling and enhanced prioritized replay with adaptive importance sampling, the proposed model achieves both efficiency and robustness. This provides a reliable framework for multi-UAV collaboration in large-scale, dynamic, and resource-constrained post-disaster environments.

Experiments

In this section, simulation experiments of the proposed algorithm are conducted around two typical scenarios: post-disaster response search and dynamic containment tasks, to verify its performance in multi-UAV cooperative tasks. All the experiments are implemented based on the PyTorch framework, and the training and testing are done on NVIDIA RTX 3090 GPUs.

Table 1 Parameter settings

Hyperparameters	Values
Max episode	10,000
Memory size	150,000
Batch size	128
Soft update factor τ	0.005
Discount factor γ	0.95
Learning rate	0.0003
Smoothness control factor λ	0.6
Adaptive importance sampling factor β	$0.4 \rightarrow 1.0$

The simulation environment is defined as a two-dimensional square area of size 100×100 units, which serves as an abstract representation of the post-disaster field. Multiple UAVs are initially deployed at random positions with uniformly sampled headings. Obstacles and dynamic targets are randomly generated at the beginning of each episode to introduce variability and ensure robustness of evaluation. Each UAV has a limited observation range, which includes the relative positions and velocities of nearby UAVs and obstacles, reflecting the partially observable nature of the environment. The action space is a two-dimensional continuous control domain, representing the UAV's accelerations along the x-axis and y-axis at each step. An episode terminates when all targets are reached, a collision occurs, or the maximum step limit of 100 is exceeded.

Performance is evaluated using three primary metrics: (1) success rate, defined as the ratio of successful episodes to total episodes; (2) average episode reward, reflecting the overall task performance; and (3) average episode length, indicating task efficiency. The main training hyperparameters are listed in Table 1.

As illustrated in Fig. 2, the sensitivity analysis indicates that the proposed method is relatively robust to variations in γ and β. In particular, $\gamma = 0.95$ and $\beta = 0.4$ consistently yield the highest success rates across both scenarios, suggesting that these values provide a good balance between stability and performance.

Comparison experiments

The reward curves are illustrated in Fig. 3, where the solid line indicates the average of multiple results and the shaded areas corresponding to the curves in the figure indicate the standard deviation ranges to reflect the volatility of the results. All reward curves are smoothed to enhance visualization and readability of trends (window_size = 200).

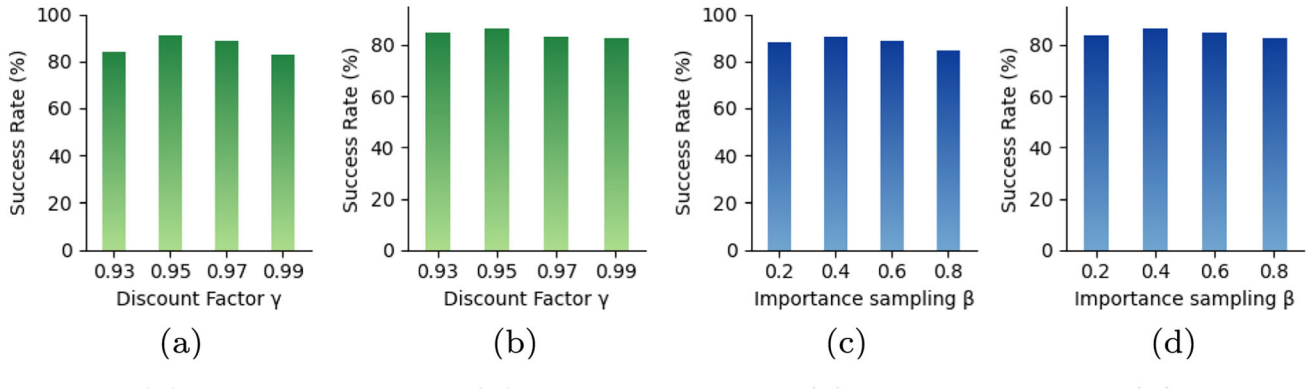


Fig. 2 Effect of γ and β on task success rate: **a** γ in post-disaster response; **b** γ in dynamic containment; **c** β in post-disaster response search; **d** β in dynamic containment

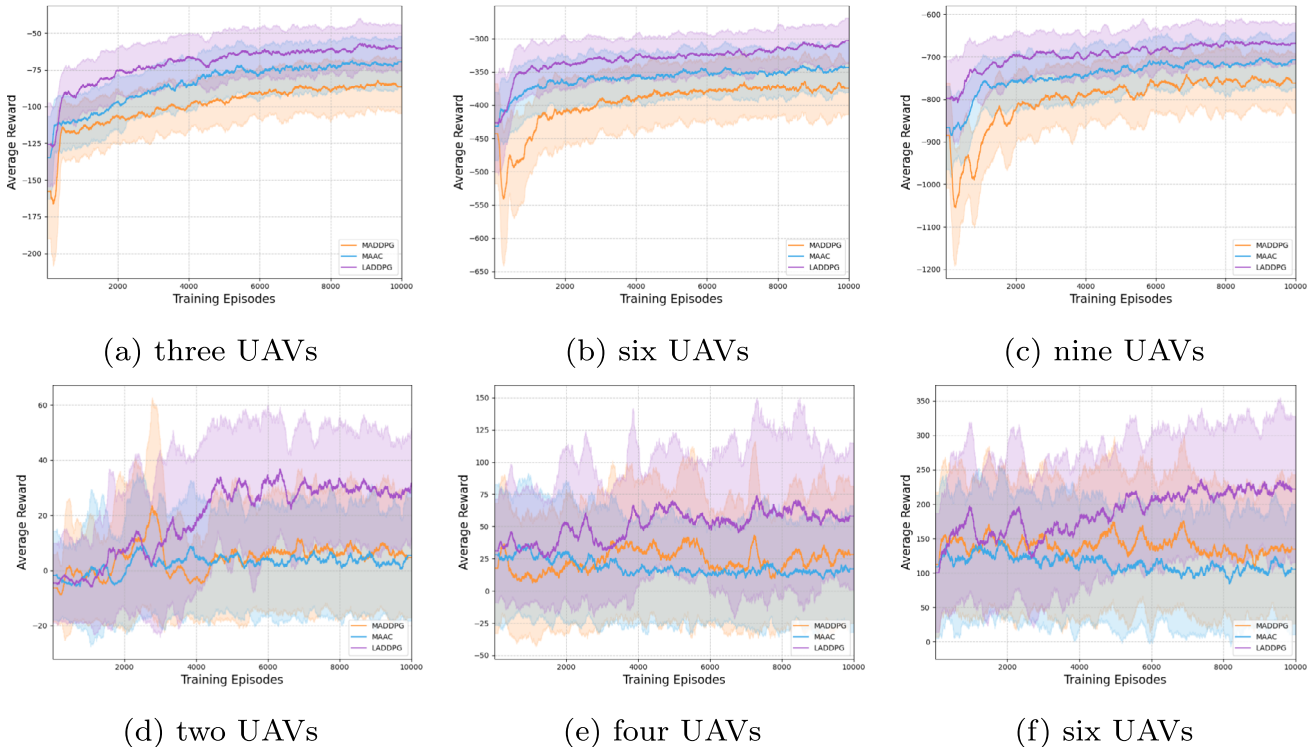


Fig. 3 Average reward curves of the post-disaster response search task (**a**–**c**) and the dynamic containment tasks (**d**–**f**)

In both post-disaster response search and dynamic containment tasks, LADDPG demonstrated significant advantages in success rate, average steps, and execution time. As shown in Fig. 3, across different drone scales, LADDPG's reward curve converged faster with less fluctuation, whereas MADDPG and MAAC exhibited substantial volatility. As the number of UAVs increases, the overall task difficulty rises, causing the success rates of all algorithms to decline. However, LADDPG exhibits the smallest decrease. For instance, Table 2 shows that in a post-disaster response search task with 9 UAVs, LADDPG maintains a success rate above 85%, while MADDPG achieves only 64%. Additionally, LADDPG demonstrates significantly lower average steps and completion times. Similarly, itn the dynamic containment task (Table 3), LADDPG maintained its lead in both success rate and efficiency metrics.

From the perspective of training overhead, Fig. 4 shows that MADDPG requires the least time due to its simpler model structure, but its performafnce is limited. MAAC achieves some performance improvement by introducing a softmax attention mechanism, but its quadratic complexity leads to a significant increase in training time. In contrast, LADDPG employs a linear attention mechanism, reducing computational burden while achieving a balance between

Table 2 Success rate of the post-disaster response search task

Number	Algorithm	Success rate (%)	Step	Time
3UAVs	MADDPG	99.0	15.53	0.03
	MAAC	99.5	13.67	0.03
	LADDPG	99.7	12.93	0.02
6UAVs	MADDPG	73.3	35.42	0.23
	MAAC	83.5	32.92	0.15
	LADDPG	90.6	20.40	0.09
9UAVs	MADDPG	64.3	56.13	0.47
	MAAC	76.1	42.79	0.32
	LADDPG	85.2	30.26	0.21

Table 3 Success rate of the dynamic containment tasks

Number	Algorithm	Success rate (%)	Step	Time
2UAVs	MADDPG	78.6	42.38	0.39
	MAAC	86.2	38.61	0.31
	LADDPG	90.1	31.22	0.20
4UAVs	MADDPG	70.3	48.60	0.31
	MAAC	81.1	58.21	0.36
	LADDPG	86.5	46.05	0.26
6UAVs	MADDPG	64.3	48.60	0.37
	MAAC	71.9	32.60	0.36
	LADDPG	80.2	39.55	0.27

performance and efficiency. Specifically, its training time is approximately 30–40% longer than MADDPG's but still nearly half that of MAAC. Notably, LADDPG's inference cost is comparable to MADDPG's, indicating this trade-off has limited impact on real-time deployment.

We further compared Transformer-based MADDPG and LADDPG, as shown in Fig 5. LADDPG outperformed Trans-MADDPG in both tasks. In the post-disaster response search task, it converged faster and exhibited higher stability. In the dynamic containment task, although slightly weaker initially, it ultimately performed better. This primarily stems from the transformer's tendency to model global dependencies, which can lead to slow convergence during high-frequency interactions. In contrast, LADDPG utilizes a linear attention mechanism to extract key features more efficiently, thereby enhancing convergence speed and stability.

The results in Table 4 further support this conclusion. Compared to trans-MADDPG, LADDPG achieves a 5–7% improvement in success rates for both post-disaster response search and dynamic containment tasks, while significantly reducing average steps and time. This demonstrates that LADDPG not only converges more efficiently during training but also enables faster and more stable collaboration at the execution level, highlighting its advantages in multi-UAV cooperative scenarios.

Therefore, the stability and efficiency of LADDPG in multi-UAV collaborative missions can be confirmed, highlighting its application potential in disaster relief scenarios requiring rapid response and operating under resource constraints.

Simulation experiment

Through simulation experiments, we analyzed the experimental scenarios and the corresponding attention distribution. The color of each UAV in the left graph corresponds to the color in the right bar graph. It can be observed that all UAVs have the highest attentional weights for themselves.

Figure 6a shows the simulation scenario of six UAVs in the post-disaster response search task. When the environment is in the state shown in this figure, UAV2 and UAV3 are in close proximity to each other, and in order to avoid collision, they assign higher attentional weights to each other.

Figure 6b shows the simulation scenario of the four UAVs in the dynamic roundup task. The green circles represent the dynamic signal source. In the current environment, UAV1 is the closest to the target of the dynamic signal source, so it pays the most attention to itself; while UAV2 and UAV3 are close to each other and far away from the signal source, so they assign relatively high attention to each other.

Ablation experiment

To verify the effectiveness of each component, we conduct ablation experiments in a post-disaster response search task with three UAVs. In order to get a more detailed view of the early stage of the model training, we selected the data of the first 1000 training sets for visualization, and all curves were smoothed (window size = 50). The experiment uses MADDPG as the baseline model, and introduces linear attention (MADDPG+RFA), smoothed and adaptive prioritized experience replay (MADDPG+SPER) and LADDPG for comparison.

As shown in Fig. 7, MADDPG exhibits significant fluctuations during the early training phase and converges noticeably slower. After incorporating SPER, the model's curve becomes more stable in the initial stages, and convergence speed improves, indicating that the adaptive experience replay mechanism effectively mitigates instability caused by uneven sample distribution. In contrast, MADDPG+RFA achieves higher average rewards after stabilization, demonstrating the advantage of linear attention in capturing critical information between agents. The final model, LADDPG, integrates both improvements, not only rapidly increasing reward values in the early stages but also maintaining minimal fluctuations and optimal stability throughout subsequent training.

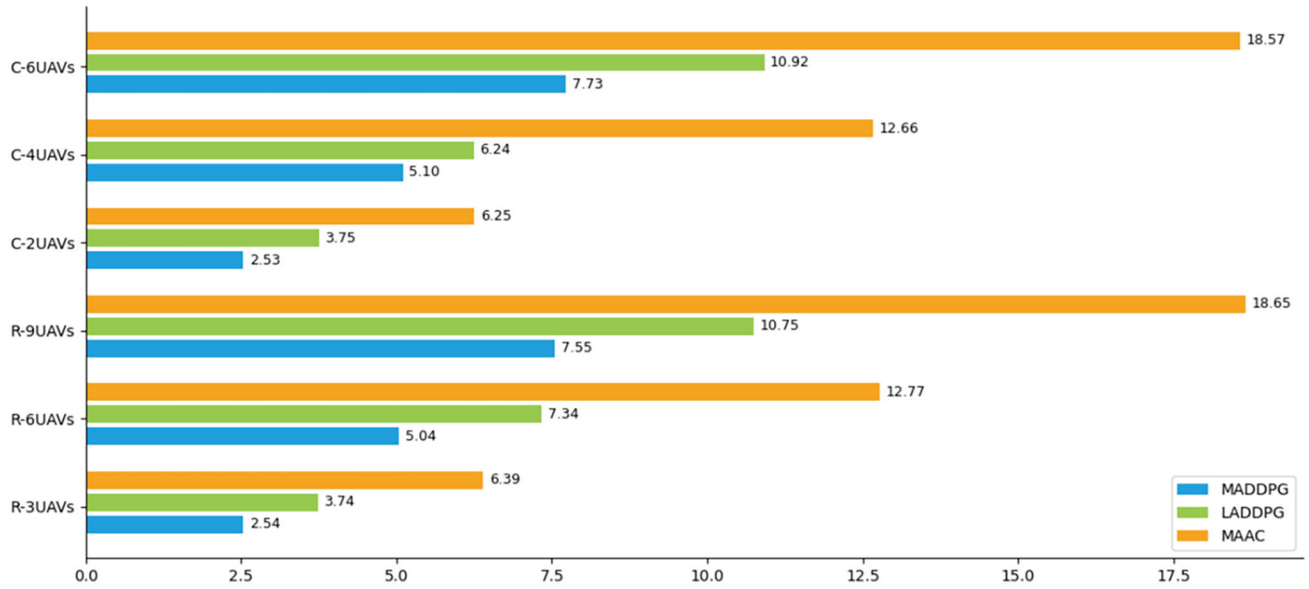


Fig. 4 Wall time for all cases

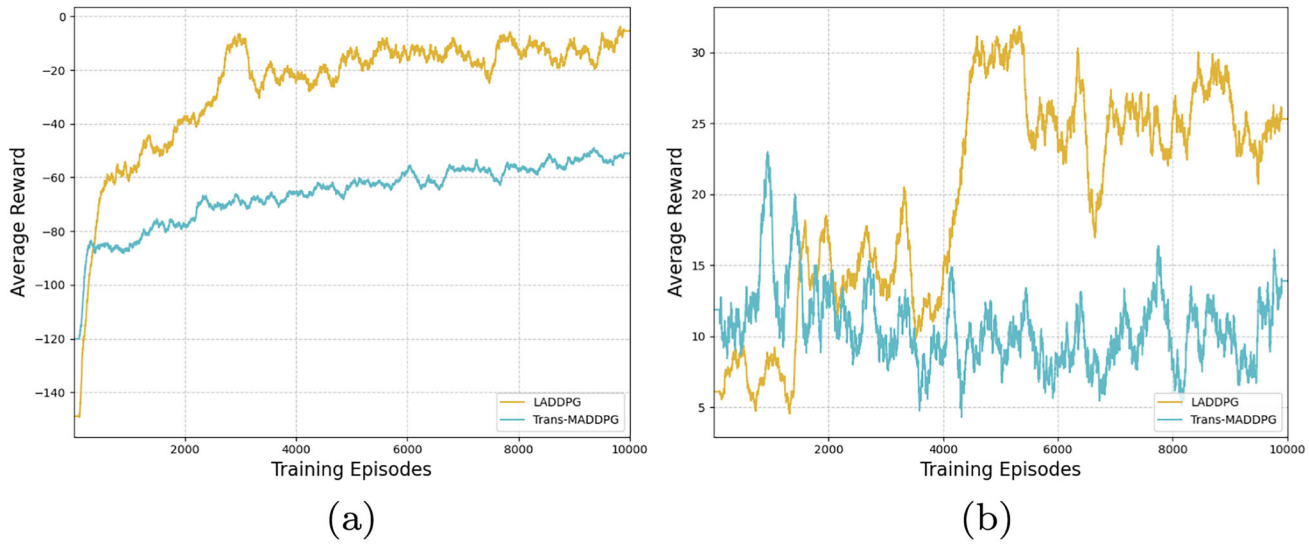


Fig. 5 Average reward comparison between Trans-MADDPG and LADDPG. **a** Post-disaster response search task with 6 UAVs. **b** Dynamic containment tasks with 4 UAVs

Table 4 Performance comparison between LADDPG and trans-MADDPG

	Post-disaster response search		Dynamic containment tasks	
	Trans-MADDPG	LADDPG	Trans-MADDPG	LADDPG
Success rate (%)	83.20	90.60	80.90	86.50
Step	31.25	20.40	60.11	46.05
Time	0.16	0.09	0.35	0.26

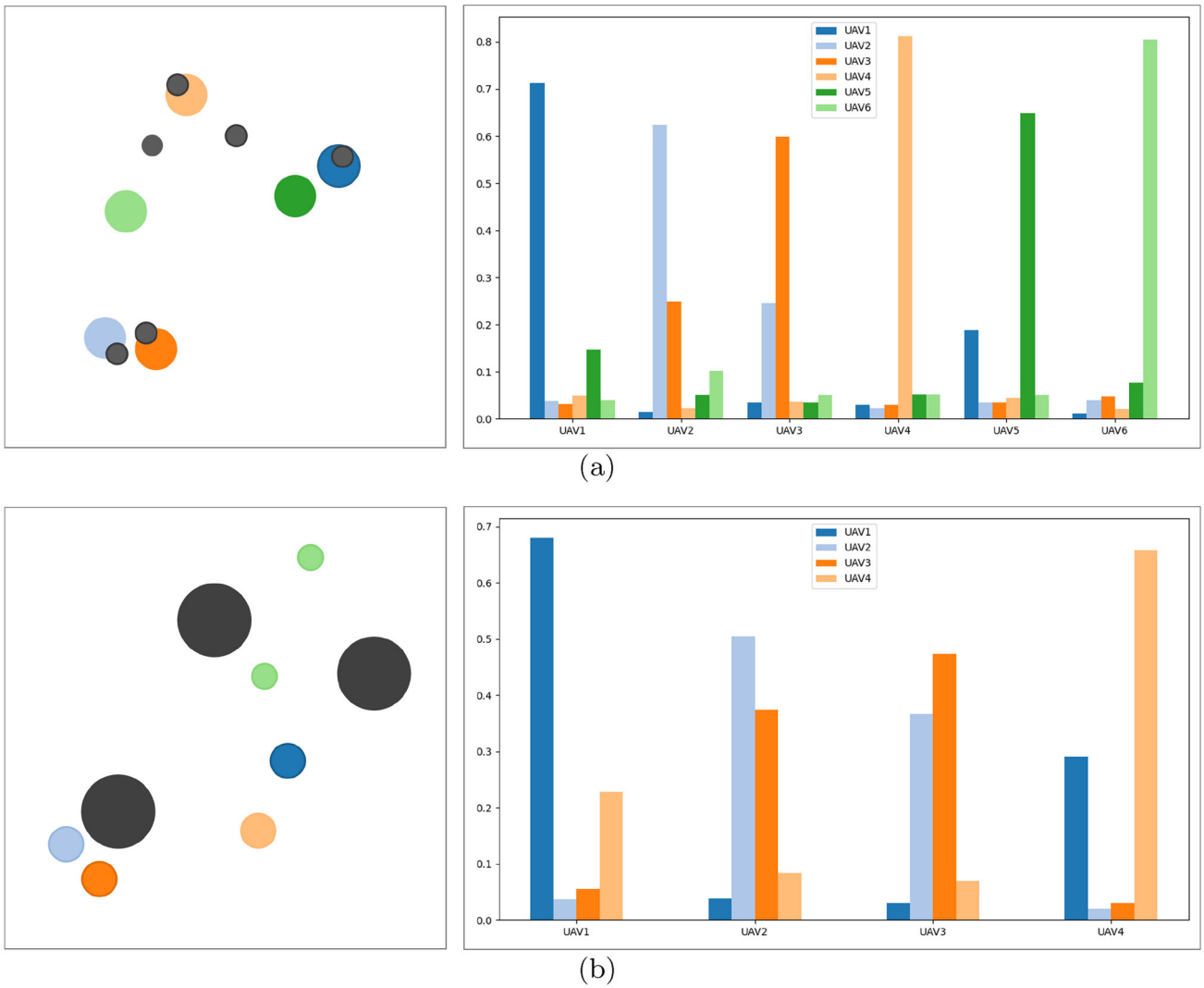


Fig. 6 Simulation experiment and its attention allocation

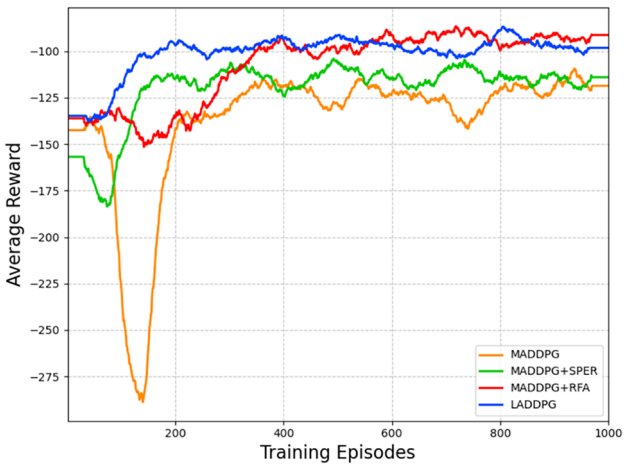


Fig. 7 Average reward for ablation experiments

These results demonstrate that SPER accelerates early convergence, RFA enhances policy quality, and their combination maximizes model stability and convergence speed, thereby validating the effectiveness and necessity of each component in LADDPG.

Conclusion

In this paper, we propose an efficient cooperative method for multi-UAV systems, aimed at addressing the challenge of rapid response in post-disaster search and rescue missions. By introducing a linear attention mechanism, the convergence speed of the model is significantly improved, while the computational complexity is effectively reduced compared to the traditional attention mechanism, thus improving the overall computational efficiency. Simulation results demonstrate

that the proposed method achieves a higher task completion rate in less time, validating its potential for application in post-disaster rescue scenarios. Future work will explore the applicability of this approach in complex three-dimensional environments.

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Author Contributions Li Tan: Proposed the research direction, provided guidance, and reviewed the manuscript. Haixia Zhao: Conducted the main research work, including algorithm design, experiments, and writing.

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